

ESSE ASSIGNMENT TECHNICAL REPORT ENVIRONMENTAL POLLUTION REPORTING

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ABSTRACT

Environmental pollution is one of the major challenges affecting human health, ecosystems and sustainable development. Pollution can occur in many forms, including air pollution, water pollution, soil contamination, noise pollution and improper waste disposal. Although government agencies and local authorities have systems for receiving environmental complaints, reporting can often be fragmented, difficult to track and slow to convert into actionable information.

This report proposes **EcoAlert**, a web-based Environmental Pollution Reporting and Monitoring System designed to provide a centralized platform for citizens to report pollution incidents. The proposed system allows users to submit complaints with photographs, descriptions, pollution categories and geographical locations. The system uses location information and a severity-based classification mechanism to prioritize complaints. An administrative dashboard enables authorities to verify complaints, assign them to responsible departments, update their status and monitor pollution hotspots.

The proposed solution supports **SDG 11** – Sustainable Cities and Communities, while also contributing to **SDG 3**-Good health and well-being, **SDG 6**- Clean water and sanitation and **SDG 12**-Responsible consumption and production. The system is designed according to environmental management principles associated with ISO 14001 and can support data-driven environmental decision-making. A basic life-cycle perspective is also considered to evaluate the environmental impacts of developing and operating the digital platform.

EcoAlert can improve public participation, transparency, response coordination and environmental monitoring while requiring relatively low infrastructure compared with physical pollution-control projects.

INTRODUCTION

Environmental pollution occurs when harmful substances or forms of energy are introduced into the natural environment at levels that negatively affect humans, animals, plants or ecosystems. Major forms include air, water, soil and noise pollution. Urbanization, industrialization, transportation, construction, improper waste disposal and other human activities can increase pollution levels.

Pollution is not only an environmental problem but also a public-health and economic problem. According to the United Nations Environment Programme (UNEP), environmental contamination contributes to approximately nine million deaths annually, while air pollution alone is responsible for nearly seven million deaths. The scale of pollution also makes effective monitoring important. UNEP's 2024 assessment describes pollution and waste as part of the world's broader triple planetary crisis, together with climate change and biodiversity loss. In India, air pollution is particularly important in urban areas. The Government of India reported that the National Clean Air Programme (NCAP) covers 131 cities across 24 States/Union Territories. The programme's target was revised to achieving up to a 40% reduction in PM₁₀ concentrations or meeting the national standard of 60 µg/m³ by 2025–26. The government also reported that 95 of the 131 NCAP cities showed improvement in annual PM₁₀ concentrations during FY 2023–24 compared with the 2017–18 baseline, while 18 cities met the national PM₁₀ standard. These statistics demonstrate the importance of continuous environmental monitoring and public participation. A resident who observes illegal waste burning, sewage discharge, excessive industrial smoke or unauthorized dumping should have a convenient way to report the incident.

Therefore, technology can be used to bridge the gap between citizens, environmental information and authorities.

PROBLEM STATEMENT

Existing Problem

Environmental pollution incidents occur regularly in residential areas, cities and industrial regions. Examples include:

- Open burning of waste
- Smoke from industries or vehicles
- Illegal dumping of solid waste
- Discharge of untreated wastewater
- Burning of plastics
- Unauthorized disposal of hazardous materials

A major problem is that citizens may not know where or how to report such incidents. Even when complaint systems exist, information may be distributed across different departments and platforms. Another issue is that a complaint submitted without accurate location, evidence or classification may be difficult to investigate.

Existing limitations

1. Lack of a centralized reporting platform.
2. Difficulty in attaching accurate geographical information.
3. Lack of systematic pollution categorization.
4. Limited visibility of complaint progress.
5. Difficulty identifying pollution hotspots.

India already uses public grievance mechanisms in environmental programmes. For example, the government has reported that public grievance redressal systems/helplines have been developed in NCAP cities. Therefore, the proposed system should not simply provide another complaint form. It should combine reporting, geographical information, evidence, prioritization, tracking and environmental analytics into one system.

SUSTAINABILITY ANALYSIS

1. Relevant Sustainable Development Goals

SDG 11 – Sustainable Cities and Communities

The primary SDG associated with this project is **SDG 11**.

Sustainable cities require effective management of environmental risks and pollution. A pollution-reporting platform supports this objective by enabling citizens to report environmental problems and helping authorities identify areas requiring intervention. The system particularly supports the objective of improving urban environmental conditions through better information, monitoring and response.

SDG 3 – Good Health and Well-being

Pollution can directly and indirectly affect human health. Air pollution, contaminated water, hazardous waste and excessive noise can contribute to health risks. EcoAlert can help identify environmental hazards earlier, allowing authorities to respond before exposure becomes more severe.

SDG 6 – Clean Water and Sanitation

Users can report:

- Sewage discharge
- Polluted rivers
- Contaminated lakes
- Illegal wastewater disposal
- Blocked or polluted drains

SDG 12 – Responsible Consumption and Production

The system can be used to report illegal dumping, excessive waste generation and improper disposal practices. This supports responsible resource and waste management.

2. ISO 14001 Analysis

ISO 14001 provides requirements for environmental management systems. ISO describes its standards as internationally agreed frameworks for managing processes, products and services. EcoAlert can support an ISO 14001-oriented environmental management approach through:

Environmental Policy: The organization implementing the platform can establish a policy focused on pollution prevention, compliance and continuous environmental improvement.

Planning: Reported pollution incidents can be used to identify environmental aspects and risks.

Environmental Aspect	Possible Impact
Waste dumping,Industrial discharge	Soil and water pollution
Vehicle smoke,Burning	Air pollution
Noise	Health disturbance

Implementation: The platform provides a mechanism through which identified environmental problems can be communicated to responsible authorities.

Monitoring: The dashboard can monitor - Number of complaints,Complaint categories ,Response time Resolution rate ,Pollution hotspots ,Repeated incidents etc.

Continual Improvement: Historical complaint data can help authorities identify recurring sources and improve environmental management strategies.

Thus, EcoAlert can function as an information and monitoring component supporting an environmental management system. ISO notes that it develops standards including ISO 14001, while certification itself is handled through conformity-assessment processes.

PROPOSED ENGINEERING/IT SOLUTION

1. EcoAlert System

EcoAlert – Smart Environmental Pollution Reporting and Monitoring System is a web-based application that enables citizens to report pollution incidents and helps authorities monitor and resolve complaints efficiently.

The system has two main interfaces:

Citizen Interface: Citizens can register/login, report pollution, select a category, upload photographs, add descriptions, submit complaints, view nearby reports, and provide feedback after resolution.

Authority Interface: Authorized officials can view and verify complaints, assign them to the concerned department, identify high-risk areas, and monitor response and resolution times.

2. Pollution Categories: The system classifies complaints into:

1. Air Pollution
2. Water Pollution
3. Soil Pollution
4. Noise Pollution
5. Plastic Pollution
6. Construction Pollution
7. Other Environmental Issues

3. Complaint Submission Process

The complaint workflow is:

Pollution observed → Complaint submitted with category, location, description and evidence → Authority verification → Complaint assigned → Action taken → Complaint resolved → Citizen feedback

This process maintains a digital record of each environmental incident from reporting to resolution.

4. Unique Features of EcoAlert

- **GPS-Based Reporting:** Records the location of reported pollution incidents.
- **Complaint Tracking:** Allows citizens to monitor complaint status.
- **Pollution Analytics Dashboard:** Provides insights into pollution trends and affected areas.
- **Smart Complaint Prioritization:** Ranks complaints based on severity, location, and environmental impact.
- **Pollution Hotspot Prediction:** Uses previous reports to identify areas likely to experience recurring pollution.

FEASIBILITY ANALYSIS

1. Technical Feasibility

EcoAlert is technically feasible as it can be developed using commonly available technologies such as HTML, CSS, JavaScript, Java, MySQL , GIS/map services, and cloud hosting. The basic system does not require specialized hardware.

2. Economic Feasibility

The prototype can be developed using open-source software and student-level development tools, keeping initial costs low. The main expenses would include web hosting, domain registration, cloud storage, database services, and security.

3. Operational Feasibility

EcoAlert is simple to use. Citizens can select a pollution type, enter details, upload evidence, and submit a report. Authorities can manage complaints through a dashboard that organizes reports based on priority.

4. Social Feasibility

The system encourages citizens to actively participate in environmental protection. By reporting pollution incidents, citizens contribute to environmental monitoring, increasing awareness and community participation.

5. Legal and Ethical Feasibility

EcoAlert must protect user privacy and prevent misuse of the platform. Key measures include secure authentication, password encryption, access control, complaint verification, personal data protection, and appropriate data retention policies. Unnecessary personal information of complainants should not be publicly displayed.

EXPECTED ENVIRONMENTAL BENEFITS

1. **Faster Pollution Identification** – Enables quick reporting and identification of pollution incidents.
2. **Better Pollution Monitoring** – Improves continuous tracking and management of pollution.
3. **Identification of Pollution Hotspots** – Helps identify areas with frequent or severe pollution.
4. **Improved Waste Management** – Supports faster reporting and better management of waste-related issues.
5. **Protection of Water Bodies** – Enables timely reporting of pollution affecting rivers, lakes, and other water bodies.
6. **Reduced Response Time** – Helps authorities respond to complaints more quickly.
7. **Increased Public Awareness** – Encourages citizens to participate in environmental protection.

JUSTIFICATION OF THE PROPOSED SOLUTION

EcoAlert is justified because pollution management requires not only regulations and monitoring equipment but also effective communication between the public and authorities. A citizen may observe pollution before an official inspection team does. Therefore, citizen-generated reports can complement conventional monitoring systems. This concept is consistent with India's broader approach to air-quality management. Under NCAP, the government has emphasized public grievance and response systems alongside air-quality monitoring and city action plans.

The proposed platform improves on a basic complaint portal by integrating:

- Geographical location
- Evidence
- Complaint tracking
- Pollution heat maps
- Analytics

Therefore, EcoAlert is not simply a website for submitting complaints. It is a digital environmental monitoring and decision-support system.

The system also has strong scalability. A small municipality could initially use it for waste dumping and air pollution. Later, it could be expanded to include water-quality sensors, IoT devices, satellite information and automated pollution alerts.

CONCLUSION

Environmental pollution is a complex problem that requires continuous monitoring, public participation and effective administrative action. Existing environmental programmes demonstrate the importance of monitoring pollution and responding to public complaints. However, fragmented reporting and limited data integration can reduce the effectiveness of environmental response.

This report proposed EcoAlert – Smart Environmental Pollution Reporting and Monitoring System, a web-based platform that integrates pollution reporting, location tracking, evidence collection, severity classification, complaint management and environmental analytics.

The system primarily supports SDG 11 – Sustainable Cities and Communities, while also contributing to SDGs 3, 6 and 12. It can support the principles of environmental management associated with ISO 14001 by improving environmental monitoring, documentation and continual improvement.

The project is technically, economically and operationally feasible because it can be developed using commonly available web technologies and relatively low-cost infrastructure. Its greatest potential benefit is the creation of a structured connection between citizens who observe environmental problems and authorities responsible for addressing them.

Therefore, EcoAlert can serve as a practical IT-based solution for improving environmental governance, increasing public participation and supporting cleaner and more sustainable communities.

REFERENCES

[1] United Nations Environment Programme, “Pollution and health,” UNEP, 2024.

[2] United Nations Environment Programme, *Global Waste Management Outlook 2024*, UNEP, 2024. The report estimates that global municipal solid waste generation was about 2.1 billion tonnes in 2023 and could reach 3.8 billion tonnes by 2050.

[3] United Nations Environment Programme, *Global Resources Outlook 2024*, UNEP, 2024.

[4] World Health Organization, “Air pollution,” WHO, 2026.

[5] Ministry of Environment, Forest and Climate Change, Government of India, “Goals set under National Clean Air Programme,” Press Information Bureau, Aug. 8, 2024.

[6] Ministry of Environment, Forest and Climate Change, Government of India, “Deaths due to pollution in the country,” Press Information Bureau, Jul. 25, 2024.

[7] Ministry of Environment, Forest and Climate Change, Government of India, “National Clean Air Programme (NCAP) to improve air quality in 131 cities by engaging all stakeholders,” Press Information Bureau.